

Regole operazionali:

Dictionary:

$$\forall (i, v1) \in d \Rightarrow \text{DictVal}(i, \text{env} \triangleright v1 \Rightarrow v2) \in d1 \Rightarrow v$$

$$\text{env} \triangleright \text{Dictionary}(\text{list}) \Rightarrow v$$

Has_key:

$$\text{env} \triangleright \text{Dictionary}(\text{list}) \Rightarrow v1$$
$$(\exists id \in v1 \text{ t.c. } v1.i == id) \Rightarrow \text{true} \Rightarrow v \quad !(\exists id \in v1 \text{ t.c. } v1.i == id) \Rightarrow \text{false} \Rightarrow v$$

$$\text{env} \triangleright \text{Has_key}(id, \text{list}) \Rightarrow v$$

Insert:

$$\text{env} \triangleright \text{Dictionary}(\text{list}) \Rightarrow v1 \quad \text{env} \triangleright \text{val} \Rightarrow \text{val1}$$
$$!(\exists id \in v1 \text{ t.c. } v1.i == id) \Rightarrow (id, \text{val1}) \Rightarrow v$$

$$\text{env} \triangleright \text{Insert}(id, \text{val}, \text{list}) \Rightarrow v$$

Delete:

$$\text{env} \triangleright \text{Dictionary}(\text{list}) \Rightarrow v1$$
$$(\exists id \in v1 \text{ t.c. } v1.i == id) \Rightarrow \text{true} \Rightarrow !(\exists id \in v \text{ t.c. } v.i == id) \Rightarrow v$$

$$\text{env} \triangleright \text{Delete}(\text{list}, id) \Rightarrow v$$

Iterate:

$\text{env} \triangleright f \Rightarrow \text{FunVal}(\text{arg}, \text{exp}, r)$

$\forall (i, v1) \in d \Rightarrow \text{DictVal}(i, \text{env} \triangleright \text{FunCall}(f, v1)) \in d1 \Rightarrow v$

$\text{env} \triangleright \text{Iterate}(f, \text{list}) \Rightarrow v$

Fold:

$\text{env} \triangleright f \Rightarrow \text{FunVal}(\text{arg}, \text{exp}, r) \quad \text{acc} \Rightarrow \text{Int}(0)$

$\forall (i, v1) \in d \Rightarrow \text{env} \triangleright \text{FunCall}(f, v1) + \text{acc} \Rightarrow \text{acc}$

$\text{env} \triangleright \text{Fold}(f, \text{list}) \Rightarrow \text{acc}$

Filter:

$\text{env} \triangleright \text{Dictionary}(\text{list}) \Rightarrow d$

$\forall (i, v1) \in d \ \&\& \ \exists \text{kl.id} == i \Rightarrow \text{DictVal}(i, v1) \in d1 \Rightarrow v$

$\text{env} \triangleright \text{Filter}(\text{kl}, \text{list}) \Rightarrow v$